WORKED SOLUTION

STEP III 1990, Question 5

Pure • Leibniz's Theorem

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q5

The key idea

Pascal's rule is just fractions with factorials, once we put both terms over a common denominator. Leibniz's formula for the nth derivative of a product is then a proof by induction on n: differentiating the nth-order formula once more produces each mixed term twice, and Pascal's rule is exactly what merges each pair. For the last part, differentiating $y = \sin^{-1} x$ directly n times is hopeless. Instead we find a simple second-order equation that y satisfies, and apply Leibniz to that.

(The printed question says $y = \sin^{-1} y$. This is clearly a misprint for $y = \sin^{-1} x$, and that is what we use.)

Solution

Pascal's rule

Recall $\binom{n}{r} = \dfrac{n!}{r!\,(n-r)!}$, with $0! = 1$. For $1 \le r \le n$ both $r-1$ and $n-r$ are non-negative, so every factorial below makes sense. The common denominator is $r!\,(n-r+1)!$, since $r! = r\,(r-1)!$ and $(n-r+1)! = (n-r+1)\,(n-r)!$:

$$\begin{aligned}\binom{n}{r} + \binom{n}{r-1} &= \frac{n!}{r!\,(n-r)!} + \frac{n!}{(r-1)!\,(n-r+1)!} \\ &= \frac{n!\,(n-r+1)}{r!\,(n-r+1)!} + \frac{n!\,r}{r!\,(n-r+1)!} \\ &= \frac{n!\,(n+1)}{r!\,(n+1-r)!} = \frac{(n+1)!}{r!\,(n+1-r)!} = \binom{n+1}{r}.\end{aligned}$$

Leibniz's formula

We assume u and v can be differentiated as often as needed. In summation notation the claim is

$$(uv)^{(n)} = \sum_{r=0}^{n} \binom{n}{r} u^{(n-r)} v^{(r)},$$

where $z^{(0)}$ means z itself. We prove this by induction on $n \geq 1$.

Base case $n = 1$**.** The right-hand side is $u^{(1)}v + uv^{(1)}$, which is $(uv)'$ by the product rule.

Inductive step. Suppose the formula holds for some $n \geq 1$. Differentiate it, using the product rule on each term:

$$(uv)^{(n+1)} = \sum_{r=0}^{n} \binom{n}{r} u^{(n+1-r)} v^{(r)} + \sum_{r=0}^{n} \binom{n}{r} u^{(n-r)} v^{(r+1)}.$$

In the second sum, put $s = r + 1$, so it becomes $\sum_{s=1}^{n+1} \binom{n}{s-1} u^{(n+1-s)} v^{(s)}$. Now both sums are made of terms $u^{(n+1-r)}v^{(r)}$. Split off the $r = 0$ term of the first sum and the $s = n + 1$ term of the second:

$$\begin{aligned}(uv)^{(n+1)} &= u^{(n+1)}v + \sum_{r=1}^{n} \left[\binom{n}{r} + \binom{n}{r-1} \right] u^{(n+1-r)} v^{(r)} + uv^{(n+1)} \\ &= u^{(n+1)}v + \sum_{r=1}^{n} \binom{n+1}{r} u^{(n+1-r)} v^{(r)} + uv^{(n+1)},\end{aligned}$$

by Pascal's rule, which applies because $1 \leq r \leq n$. Since $\binom{n+1}{0} = \binom{n+1}{n+1} = 1$, this is

$$(uv)^{(n+1)} = \sum_{r=0}^{n+1} \binom{n+1}{r} u^{(n+1-r)} v^{(r)},$$

the formula for $n + 1$. By induction it holds for every $n \geq 1$.

The derivatives of $\sin^{-1} x$

Let $y = \sin^{-1} x$ for $-1 < x < 1$. Then

$$y^{(1)} = \frac{1}{\sqrt{1 - x^2}}, \qquad \text{so} \qquad (1 - x^2)(y^{(1)})^2 = 1.$$

Differentiating this with respect to x,

$$(1 - x^2) \cdot 2y^{(1)} y^{(2)} - 2x(y^{(1)})^2 = 0.$$

Since $y^{(1)} > 0$ on $(-1, 1)$, we may divide by $2y^{(1)}$:

$$(1 - x^2)\, y^{(2)} - x\, y^{(1)} = 0. \qquad (*)$$

This is the required result for $n = 0$.

Ask yourself Why square $y^{(1)}$ before differentiating, rather than just differentiating $(1-x^2)^{-1/2}$?

Differentiating $(1-x^2)^{-1/2}$ directly gives $y^{(2)} = x(1-x^2)^{-3/2}$, and multiplying by $1-x^2$ gives $(*)$ just as well. Either way works. The point is to reach an equation whose coefficients are *polynomials* in x and which is *linear* in the derivatives of y, because that is what Leibniz handles neatly: differentiating a polynomial coefficient soon gives zero.

Now differentiate $(*)$ n times, for $n \geq 1$, applying Leibniz's formula to each product. For the first, take $u = y^{(2)}$ and $v = 1-x^2$. Then $v^{(1)} = -2x$, $v^{(2)} = -2$ and all higher derivatives vanish, so only three terms survive:

$$\left((1-x^2)\,y^{(2)}\right)^{(n)} = (1-x^2)\,y^{(n+2)} + \binom{n}{1}(-2x)\,y^{(n+1)} + \binom{n}{2}(-2)\,y^{(n)}.$$

(When $n = 1$ the last term is absent, which agrees with $\binom{1}{2} = 0$.) For the second, take $u = y^{(1)}$ and $v = x$, so only two terms survive:

$$\left(x\,y^{(1)}\right)^{(n)} = x\,y^{(n+1)} + n\,y^{(n)}.$$

Subtracting, the nth derivative of $(*)$ is

$$(1-x^2)\,y^{(n+2)} - 2nx\,y^{(n+1)} - n(n-1)\,y^{(n)} - x\,y^{(n+1)} - n\,y^{(n)} = 0.$$

Collecting terms, $-2nx - x = -(2n+1)x$ and $-n(n-1) - n = -n^2$, so

$$(1-x^2)\,y^{(n+2)} - (2n+1)x\,y^{(n+1)} - n^2 y^{(n)} = 0,$$

as required. It holds for all $n \geq 0$ and all x in $(-1,1)$.

Checks

- Take $n = 1$ in Leibniz: we get the product rule. With $n = 2$: $(uv)'' = u''v + 2u'v' + uv''$, which you can confirm by differentiating $u'v + uv'$.
- At $x = 0$ the final result says $y^{(n+2)}(0) = n^2 y^{(n)}(0)$. With $y^{(1)}(0) = 1$ this gives $y^{(3)}(0) = 1$ and $y^{(5)}(0) = 9$, so the Maclaurin series begins $x + \frac{x^3}{6} + \frac{9x^5}{120} = x + \frac{x^3}{6} + \frac{3x^5}{40}$, which is the known series for $\sin^{-1} x$.
- We checked numerically: Pascal's rule for all $n < 60$; Leibniz exactly on 300 random pairs of polynomials; and the final identity for n up to 27 at six points in $(-1,1)$, using Taylor coefficients of $(1-x^2)^{-1/2}$. The relative residual was below 10^{-15}.

Common slips

- Getting the binomial coefficient wrong after reindexing the second sum in the inductive step. Write out the $r = 0$ and $r = n+1$ end terms separately before applying Pascal's rule.

- Trying to differentiate $\sin^{-1} x$ n times directly, instead of first finding the equation $(*)$.
- Forgetting the factor $\binom{n}{2}$, or the minus signs, when differentiating $1 - x^2$.

Beyond STEP

The trick of finding a linear differential equation with polynomial coefficients and then differentiating it n times is a standard way to get recurrences for Taylor coefficients. It leads directly to the method of power-series solutions of ODEs, which is how Legendre and Bessel functions arise. Leibniz's formula is also the calculus version of the binomial theorem: if $u = \mathrm{e}^{ax}$ and $v = \mathrm{e}^{bx}$, it reduces to $(a+b)^n = \sum \binom{n}{r} a^{n-r} b^r$. For more on series solutions, see Boyce and DiPrima, *Elementary Differential Equations*.

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